$$f(x) = \frac{1}{\sqrt{|x^2 - x| + x}}$$

Determinare l'insieme di derivabilità della funzione e scrivere le equazioni delle rette tangenti nei punti di non derivabilità di $f(x)$

D: $|x^2 - x| + x > 0$

1) $x^2 - x \geq 0 \quad x = 0 \quad x = 1$

$x \leq 0 \vee x \geq 1$

$x^2 - x + x > 0 \quad \forall x \neq 0$

$\Rightarrow \boxed{x < 0 \vee x \geq 1}$

2) $0 < x < 1$

$-x^2 + x + x > 0$

$-x^2 + 2x > 0 \quad x = 0 \quad x = 2$

$0 < x < 2$

$\Rightarrow \boxed{0 < x < 1}$

0 1 2

$\Rightarrow$ D: $\mathbb{R} \setminus \{0\}$

$$f(x) = \begin{cases} \dfrac{1}{\sqrt{x^2}} & x < 0 \vee x \geq 1 \\ \dfrac{1}{\sqrt{-x^2 + 2x}} & 0 < x < 1 \end{cases}$$

$$f(x)=\begin{cases}\frac{1}{|x|} & x<0 \vee x\geq 1\\ \frac{1}{\sqrt{-x^2+2x}} & 0<x<1\end{cases}$$

$$f(x)=\begin{cases}\frac{1}{x} & x\geq 1 \rightarrow f(1)=1\\ -\frac{1}{x} & x<0\\ \frac{1}{\sqrt{-x^2+2x}} & 0<x<1\end{cases}$$

Studio la derivabilità in $x_0=1$

$$\lim_{h\to 0}\frac{f(x_0+h)-f(x_0)}{h}$$

$$\lim_{h\to 0^-}\frac{\frac{1}{\sqrt{-(1+h)^2+2(1+h)}}-1}{h}$$

$$\lim_{h\to 0^-}\frac{\frac{1}{\sqrt{-1-2h-h^2+2+2h}}-1}{h}$$

$$\lim_{h\to 0^-}\frac{\frac{1}{\sqrt{1-h^2}}-1}{h}=\lim_{h\to 0^-}\frac{1-\sqrt{1-h^2}}{h\sqrt{1-h^2}}$$

$$\lim_{h\to 0^-} \frac{-\left(\sqrt{1-h^2}-1\right)}{h\sqrt{1-h^2}}$$

$(1+x)^{\alpha} - 1 \sim \alpha x$ per $x \to 0$

$$\lim_{h\to 0^-} \frac{-\left(-\frac{1}{2}h^2\right)}{h\sqrt{1-h^2}} = 0 = f'_-(1)$$

$$\lim_{h\to 0^+} \frac{\frac{1}{h+1}-1}{h} = \lim_{h\to 0^+} \frac{1-h-1}{h(h+1)} = -1 = f'_+(1)$$

$f(x)$ NON DERIVABILE in $x_0 = 1$

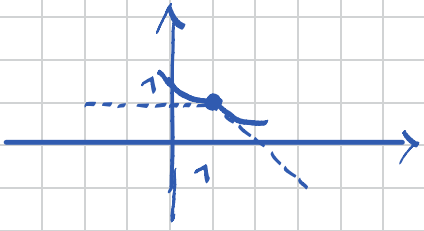
PUNTO ANGOLOSO

1) $y - y_0 = m(x - x_0)$ $\quad x_0 = 1$ $\quad y_0 = f(1) = 1$

$m = f'_-(1) \Rightarrow y - 1 = 0 \Rightarrow y = 1$

2) $m = f'_+(1) \Rightarrow y - 1 = -(x-1)$

$y = -x + 2$



Insieme di derivabilità di

$$f(x) = \begin{cases} \dfrac{\sqrt{x+1}-1}{\sqrt{x}} & x>0 \\ 0 & x=0 \end{cases}$$

D: $\mathbb{R}^+$ $[0, +\infty)$

Continuità in $x=0$

$$\lim_{x\to 0^+} \frac{\sqrt{x+1}-1}{\sqrt{x}} = \lim_{x\to 0^+} \frac{\frac{1}{2}x}{x^{1/2}} =$$

$$\lim_{x\to 0^+} \frac{1}{2}x^{1/2} = 0 = f(0) \text{ continua}$$

$$f(x) = \frac{\sqrt{x+1}-1}{\sqrt{x}}$$

$$f'(x) = \frac{\frac{1}{2\sqrt{x+1}}\cdot 1 \cdot \sqrt{x} - \frac{1}{2\sqrt{x}}\left(\sqrt{x+1}-1\right)}{x}$$

$$= \frac{\frac{x-(x+1)+\sqrt{x+1}}{2\sqrt{x+1}\cdot\sqrt{x}}}{x}$$

$$= \frac{\sqrt{x+1}-1}{2x\sqrt{x+1}\cdot\sqrt{x}}$$

D': $\begin{cases} x+1>0 \\ x>0 \\ x\neq 0 \end{cases}$

$\Rightarrow x>0$

Studio la derivabilità in $x=0$

$$\lim_{x\to 0^+} \frac{\sqrt{x+1}-1}{2x\sqrt{x+1}\cdot\sqrt{x}} =$$

PUNTO A TG. VERT.

$$\lim_{x\to 0^+} \frac{\frac{1}{2}x}{2x\sqrt{x+1}\cdot\sqrt{x}} = +\infty$$

OPPURE:

$$\lim_{h\to 0^+} \frac{\frac{\sqrt{h+1}-1}{\sqrt{h}} - 0}{h}$$

$$\lim_{h\to 0^+} \frac{\sqrt{h+1}-1}{h\sqrt{h}} = \lim_{h\to 0^+} \frac{\frac{1}{2}h}{h\sqrt{h}} = +\infty$$

$$f(x) = \arcsin\sqrt[3]{x+1}$$

$$-1 \le \sqrt[3]{x+1} \le 1$$

$$\begin{cases} \sqrt[3]{x+1} \le 1 \\ \sqrt[3]{x+1} \ge -1 \end{cases} \quad \begin{cases} x+1 \le 1 \\ x+1 \ge -1 \end{cases} \quad \begin{cases} x \le 0 \\ x \ge -2 \end{cases}$$

D: $-2 \le x \le 0$

$$f'(x) = \frac{1}{\sqrt{1-(x+1)^{2/3}}} \cdot \frac{1}{3}(x+1)^{-2/3}$$

$$f'(x) = \frac{1}{\sqrt{1-(x+1)^{2/3}}} \cdot \frac{1}{3(x+1)^{2/3}}$$

D' $\begin{cases} 1-(x+1)^{2/3} > 0 \\ x+1 \neq 0 \end{cases}$ $\begin{cases} (x+1)^{2/3} < 1 \\ x \neq -1 \end{cases}$

$\begin{cases} \sqrt[3]{(x+1)^2} < 1 \\ x \neq -1 \end{cases}$ $\begin{cases} (x+1)^2 < 1 \\ x \neq -1 \end{cases}$

$\begin{cases} x^2+2x+1 < 1 \\ x \neq -1 \end{cases}$ $\begin{cases} -2 < x < 0 \\ x \neq -1 \end{cases}$

$\Rightarrow$ punti da studiare: $x=-2$ $x=0$ $x=-1$

1) $x=0$ $\lim_{x\to 0^-} \frac{1}{\sqrt{1-(x+1)^{2/3}}} \cdot \frac{1}{3(x+1)^{2/3}} = +\infty$

PUNTO A TANG. VERTICALE

2) $x=-2$ $\lim_{x\to -2^+} \frac{1}{\sqrt{1-(x+1)^{2/3}}} \cdot \frac{1}{3(x+1)^{2/3}} = +\infty$

PUNTO A TANG. VERTICALE

3) $x=-1$ $\lim_{x\to -1^\pm} \frac{1}{\sqrt{1-(x+1)^{2/3}}} \cdot \frac{1}{3(x+1)^{2/3}} = \pm\infty$

FLESSO A TANG. VERTICALE

Stabilire per quali valori di a, b è continua e derivabile

$$f(x)=\begin{cases}(x-b)^2-2 & x\geq 0\\ a\sin x & x<0\end{cases}$$

Continuità in $x_0=0$

$\lim_{x\to 0^-} a\sin x = 0$ $\lim_{x\to 0^+} (x-b)^2-2 = b^2-2$

$f(0)=b^2-2 \Rightarrow b^2-2=0 \quad \boxed{b=\pm\sqrt{2}}$

Se $b\neq\pm\sqrt{2}$ $f(x)$ non derivabile in $x=0$

Se $b=\pm\sqrt{2}$

$b=-\sqrt{2}$ $f(x)=\begin{cases}(x+\sqrt{2})^2-2 & x\geq 0\\ a\sin x & x<0\end{cases}$

$$f'(x)=\begin{cases}2(x+\sqrt{2}) & x>0\\ a\cos x & x<0\end{cases}$$

$\lim_{x\to 0^-} a\cos x = a$

$\lim_{x\to 0^+} 2(x+\sqrt{2}) = 2\sqrt{2}$

$\Rightarrow \boxed{a=2\sqrt{2}}$

$\boxed{b=-\sqrt{2}}$

Se $b=\sqrt{2}$ $\quad f(x)=\begin{cases}(x-\sqrt{2})^2-2 & x\geq 0\\ a\sin x & x<0\end{cases}$

$$f'(x)=\begin{cases}2(x-\sqrt{2}) & x>0\\ a\cos x & x<0\end{cases}$$

$\lim_{x\to 0^+} 2(x-\sqrt{2}) = -2\sqrt{2}$

$\lim_{x\to 0^-} a\cos x = a$

$\Rightarrow \boxed{\begin{array}{l}a=-2\sqrt{2}\\ b=\sqrt{2}\end{array}}$

Sia $f(x)=e^{x^2+x+1}$

a. Calcolare $f'(x)$ e stabilire se $f(x)$ è monotona per $x>-\frac{1}{2}$

b. Detta g la funzione inversa di f nell'intervallo $x>-\frac{1}{2}$, calcolare $g'(e^3)$

$f(x) = e^{x^2+x+1}$

$f'(x) = e^{x^2+x+1}(2x+1)$

$f'(x) > 0 \Rightarrow e^{x^2+x+1}(2x+1) > 0$

$2x+1 > 0 \quad x > -\frac{1}{2}$

$\Rightarrow f(x)$ è monotona strettam. crescente

$\Rightarrow f(x)$ è invertibile in $x > -\frac{1}{2}$

$g'(y_0) = \frac{1}{f'(x_0)} \qquad y_0 = e^3$

$e^3 = e^{x^2+x+1}$

$\boxed{x_0 = 1}$

$3 = x^2+x+1$

$x^2+x-2 = 0$

$x_{1,2} = \frac{-1 \pm 3}{2} \begin{cases} -2 & \text{no} \\ 1 & (x > -\frac{1}{2}) \end{cases}$

$g'(e^3) = \frac{1}{f'(1)} = \frac{1}{3e^3}$

Sia $f(x) = \frac{x}{\log x}$

a. Calcolare $f'(x)$ e dedurre che $f(x)$ è monotona in $(e, +\infty)$ quindi invertibile

b. Detta g la funzione inversa def nell'intervallo dato, calcolare

$$g'(e^2/2)$$

D: $x > 0 \quad x \neq 1 \quad (\log x \neq 0)$

$$f'(x) = \frac{\log x - \frac{1}{x} \cdot x}{\log^2 x} = \frac{\log x - 1}{\log^2 x}$$

$$f'(x) > 0 \qquad \frac{\log x - 1}{\log^2 x} > 0$$

$N > 0 \quad \log x - 1 > 0 \quad \log x > 1 \quad x > e$

$D > 0 \quad \log^2 x > 0 \quad \forall x \neq 1$

$\Rightarrow f'(x) > 0$ in $x > e$ $(e, +\infty)$ monotona strettamente crescente

$$g'(e^2/2) \qquad y_0 = e^2/2$$

$$\boxed{x = e^2}$$

$$\frac{e^2}{2} = \frac{x}{\log x}$$

$$\log e^2 = 2 \log e = 2$$

$$g'(e^2/2) = \frac{1}{f'(e^2)}$$

$$f'(x) = \frac{\log x - 1}{\log^2 x} \qquad f'(e^2) = \frac{\log e^2 - 1}{(\log e^2)^2}$$

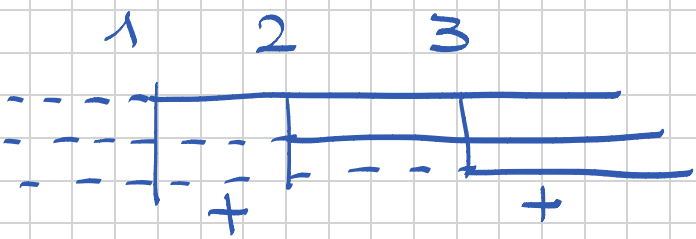
$$f'(e^2) = \frac{2-1}{4} = \frac{1}{4} \Rightarrow g'(e^2/2) = 4$$

$$f(x) = \begin{cases} \log\left[\dfrac{(x-1)(x-2)}{(x-3)}\right] & x > 3 \\ x^2\sqrt[3]{x+1} & x \le 3 \end{cases}$$

$$f_1(x) = \log\left[\frac{(x-1)(x-2)}{x-3}\right]$$

$$\frac{(x-1)(x-2)}{x-3} > 0$$

$x-1>0 \quad x>1$
$x-2>0 \quad x>2$
$x-3>0 \quad x>3$



$\Rightarrow x > 3$ va bene

D: $\mathbb{R}$

$$\lim_{x\to-\infty} x^2\sqrt[3]{x+1} = \lim_{x\to-\infty} x^2 \cdot x^{1/3} =$$

$$\lim_{x\to-\infty} x^{7/3} = -\infty$$

$$m = \lim_{x\to-\infty} \frac{f(x)}{x} = \lim_{x\to-\infty} \frac{x^2\sqrt[3]{x+1}}{x} =$$

$$\lim_{x\to-\infty} x \cdot x^{1/3} = +\infty \quad \text{non c'è AS OBL.}$$

$$\lim_{x\to+\infty} \log\left[\frac{(x-1)(x-2)}{x-3}\right] = \lim_{x\to+\infty} \log \frac{x^2}{x} = +\infty$$

$$m = \lim_{x\to+\infty} \frac{\log\left[\frac{(x-1)(x-2)}{(x-3)}\right]}{x} = 0$$

non c'è AS OBL.

Studio $x=3$

$$\lim_{x\to 3^-} x^2\sqrt[3]{x+1} = 9\sqrt[3]{4}$$ DISC. 2ª SP.

$$\lim_{x\to 3^+} \log\left[\frac{(x-1)(x-2)}{(x-3)}\right] = +\infty$$

$x=3$ A.V. DESTRO

Segno e zeri

1) $\log\left[\frac{(x-1)(x-2)}{(x-3)}\right] \geq 0$

$$\frac{(x-1)(x-2)}{(x-3)} \geq 1$$

$$\frac{x^2-x-2x+2-x+3}{x-3} \geq 0$$

$$\frac{x^2-4x+5}{x-3} \geq 0 \qquad N \geq 0 \quad x^2-4x+5 \geq 0$$

$$D > 0 \quad x > 3$$

$N > 0 \quad \forall x \qquad f(x) > 0 \quad x > 3$

2) $x^2 \sqrt[3]{x+1} \geq 0$

$x^2 \geq 0 \quad \forall x \qquad x = 0$ ~~zero~~ per $f(x)$

$\sqrt[3]{x+1} \geq 0 \qquad x \geq -1$

$\Rightarrow f(x) > 0 \quad -1 < x < 3$

$f(0) = 0$

$f(-1) = 0$

Derivata prima x^2-3x+2

1) $f(x) = \log\left[\frac{(x-1)(x-2)}{(x-3)}\right]$

$$f'(x) = \frac{(x-3)}{(x-1)(x-2)} \cdot \left[\frac{(2x-3)(x-3)-(x^2-3x+2)}{(x-3)^2}\right]$$

$$= \frac{2x^2-6x-3x+9-x^2+3x-2}{(x-1)(x-2)(x-3)} = \frac{x^2-6x+7}{(x-1)(x-2)(x-3)}$$

D': $x \neq 1 \quad x \neq 2 \quad x \neq 3$

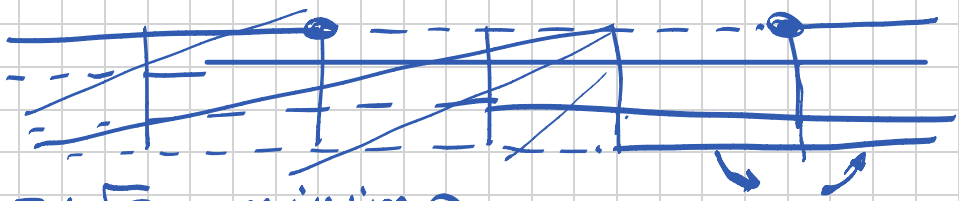
$f'(x) \geq 0 \quad \dfrac{x^2-6x+7}{(x-1)(x-2)(x-3)} \geq 0$

$N \geq 0 \quad x^2-6x+7 \geq 0$

$D > 0 \quad x>1 \quad x>2 \quad x>3$

$x_{1,2} = 3 \pm \sqrt{2} \qquad x \leq 3-\sqrt{2} \vee x \geq 3+\sqrt{2}$

$1 \quad 3-\sqrt{2} \quad 2 \quad 3 \quad 3+\sqrt{2}$



$x = 3+\sqrt{2}$ minimo

2) $f(x) = x^2 (x+1)^{1/3} \qquad x \leq 3$

$f'(x) = 2x(x+1)^{1/3} + x^2 \frac{1}{3}(x+1)^{-2/3}$

$f'(x) = 2x(x+1)^{1/3} + \dfrac{x^2}{3(x+1)^{2/3}}$

$= \dfrac{6x(x+1)+x^2}{3(x+1)^{2/3}} = \dfrac{7x^2+6x}{3(x+1)^{2/3}}$

$D': \ x \neq -1$

$\lim_{x \to -1^{\pm}} \dfrac{7x^2+6x}{3(x+1)^{2/3}} = +\infty$ Flesso a tg. vert.

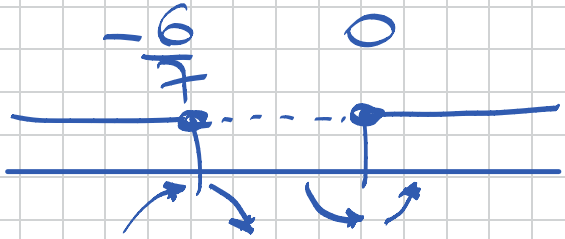
$f'(x) \geq 0 \quad \frac{7x^2+6x}{3(x+1)^{2/3}} \geq 0$

$N \geq 0 \quad 7x^2+6x \geq 0 \quad x=0 \quad x=-\frac{6}{7}$

$x \leq -\frac{6}{7} \vee x \geq 0$

$D>0 \quad (x+1)^{2/3} > 0 \quad \forall x \neq -1$

$\sqrt[3]{(x+1)^2} > 0$



max $x=-\frac{6}{7}$ min $x=0$